

# Integral, Series, Product, and Operator Representations for the Fabius–Rvachev System

A recursive corpus audit, a representation atlas, and new Fredholm,  
exterior-power, total-positivity, heat-kernel, and Mellin–Barnes bridges

Research report based on the recursive `*.tex` corpus under  
`Analysis/FabiusFunction/docs`

27 August 2026

Audit anchor reported by the repository corpus:  
32d6d36c51d803289e6d6a0dc0c37753766eba47

## Abstract

The Fabius function, Rvachev’s compactly supported up-function, their signed Thue–Morse extension, and their Fourier images admit an unusually dense web of representations: self-differential and refinement equations, random uniform series, infinite convolutions, finite box splines, signed truncated-power sums, dyadic sinc and cosine products, canonical products indexed by the 2-adic valuation, Fourier and Poisson inversion formulas, cosine and sampling series, moment and cumulant expansions, exact dyadic composition and  $q$ -binomial formulas, Laplace products, Mellin transforms, and Lambert- $W_{-1}$  endpoint coordinates. This report first organizes those formulas into a normalization-safe atlas after a recursive audit of the repository’s living references, frozen historical lineages, active frontier drafts, and transcribed papers. Results already present in the audited tree are explicitly marked as such rather than relabeled as discoveries.

The main new contribution relative to the pinned corpus is a specified operator model. Let  $G = (-d^2/dx^2)_D^{-1}$  be the Dirichlet Green operator on  $L^2(0, 1)$ , and put

$$B = \bigoplus_{n \geq 1} 4^{-n} G.$$

Then  $B$  is positive trace class and the angular Fourier transform of up is the Fredholm determinant

$$\varphi(t) = \det(I - t^2 B).$$

Equivalently,  $B^{-1}$  has eigenvalues  $(2\pi m)^2$  with multiplicity  $1 + \nu_2(m)$ . This specified operator upgrades the scalar “spectral determinant” of the existing corpus to a concrete Dirichlet-Laplacian spectrum. It yields new resolvent and heat-kernel representations, Brownian-bridge quadratic functionals, complete-Bernstein structure on the imaginary axis, and a Mellin–Barnes determinant formula for the endpoint Laplace transform.

A second new bridge identifies normalized even moments with exterior-power traces,

$$\frac{\mathbb{E} Z^{2k}}{(2k)!} = \text{Tr}(\wedge^k B),$$

where  $Z$  has density up. Combining this with the exact reciprocal-dyadic identity gives the positive spectral formula

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\text{Tr}(\wedge^k B)}{(n-2k)!}.$$

Newton identities convert it into finite combinations of  $\zeta(2j)/(\pi^{2j}(4^j - 1))$ , hence Bernoulli numbers. Moreover the sequence  $\{\mathbb{E}Z^{2k}/(2k)!\}_{k \geq 0}$  is a Pólya-frequency sequence of infinite order; therefore every Toeplitz minor is nonnegative and strict Newton–Turán inequalities follow. High-precision numerical experiments verify the product/trace identity, the exact dyadic formula, the inequalities, and the Mellin–Barnes integral. The report ends with carefully separated conjectures and a research program for Fredholm minors, operator deformations, inverse-Fabius saddles, derivative-zero statistics, and formalization.

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# 1 Scope, audit boundary, and proof-status discipline

## 1.1 What was audited

The requested source boundary is the recursive `*.tex` tree below `Analysis/FabiusFunction/docs`. The repository’s own most recent audit draft pins commit `32d6d36c51d803289e6d6a0dc0c37753766eba47` and records 67 distinct LaTeX paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (including twelve Fourier-decay dossiers), and seven transcribed or corrected papers [12]. The present study used those same lineages and compared the living primary exposition, the consolidated frontier volume, the active draft families, the archive provenance listings, and the paper-source folders. Historical revisions were treated as mathematical lineages rather than as independent claims.

The most important distinction is between two layers.

- (i) The living primary exposition records definitions and results aligned with Lean declarations: the bounded distribution function  $F$ , the compact bump up, the signed global extension, probability models, Fourier products and inversion, exact dyadic arithmetic, and the proved part of the endpoint theory.
- (ii) The non-formalized frontier volume and its descendant reports contain developed but not yet fully formalized mathematics: all-order endpoint expansions, periodic Mellin corrections, inverse asymptotics, sharp sinc-product decay, zero spectra and scalar spectral-determinant formulas, arithmetic rays, half-integer sampling, Pascal-weighted hierarchies,  $q$ -Richardson schemes, and arbitrary exponent-sequence generalizations.

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**Meaning of “new”.** Throughout this report, “new” means absent from the pinned repository corpus after path, phrase, and formula comparison, and derived or proved in this manuscript. It is not a claim of exhaustive worldwide priority. Classical ingredients are cited, and open statements are labeled as conjectures.

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## 1.2 Prior art inside the repository

The following are treated as **corpus-known** and are used as inputs rather than claimed as discoveries:

- the probability-series model, convolution structure, refinement equations, and finite Thue–Morse box-spline approximants;
- the dyadic sinc product, repeated cosine product, integer zero set, multiplicity  $1 + \nu_2(m)$ , canonical product, zero-count defect, scalar spectral zeta, scalar heat trace, local zero jets, and  $q$ -deformations;
- Fourier inversion, Poisson summation, half-integer cosine/sampling formulas, moment, cumulant, Bell-polynomial, and Bernoulli-polynomial identities;
- exact dyadic values, ordered-composition and path sums, determinant recurrences, half-base  $q$ -Pochhammer and Gaussian-binomial formulas, denominator results, and interpolation/Richardson transforms;
- endpoint Laplace products, Mellin harmonic sums, nonconstant logarithmic-periodic corrections, the lower Lambert coordinate, all-order asymptotics, and the inverse Fabius expansion;
- Thue–Morse Mahler products, Prouhet cancellation, rarefied sums, and automatic sign laws;

- arbitrary nonnegative exponent sequences and the Pascal-weighted hierarchy as scalar products and probability laws.

The phrase audit found no counterpart in the principal/current frontier sources for a specified Fredholm operator built from the Dirichlet Green kernel, Brownian-bridge quadratic functionals, exterior-power traces, Pólya-frequency/Toeplitz total positivity of normalized even moments, or the exact exterior-power formula for  $F(2^{-n})$ . One audited hierarchy report explicitly warns that its word “spectral” refers to the zero divisor and not to the spectrum of a specified operator; that gap is the starting point of [section 4](#).

## 2 Normalization dictionary and the probability model

Normalization errors are particularly easy here because three Fourier conventions and three closely related functions occur in the literature. We fix all conventions at the outset.

### 2.1 The three real functions

Let  $F: \mathbb{R} \rightarrow [0, 1]$  be the Fabius distribution function, normalized by

$$F(x) = 0 \quad (x \leq 0), \quad F(1) = 1,$$

with

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad F(1-x) = 1 - F(x) \quad (0 \leq x \leq 1). \quad (1)$$

Rvachev’s compact bump is

$$\text{up}(x) = \begin{cases} F(1 - |x|), & |x| \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (2)$$

Consequently

$$F(x) = \text{up}(x - 1) \quad (0 \leq x \leq 1), \quad F'(x) = 2 \text{up}(2x - 1). \quad (3)$$

The signed global extension is

$$\mathcal{F}(x) = \sum_{m \geq 0} (-1)^{s_2(m)} \text{up}(x - 2m - 1), \quad (4)$$

where the supports are disjoint apart from endpoints. It satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (5)$$

These identities are [corpus-known](#); see the classical constructions in [\[5, 1, 13\]](#).

### 2.2 Random uniform series

Let  $U_n$  be independent  $\text{Unif}[0, 1]$  variables and let  $V_n = 2U_n - 1$ , so  $V_n \sim \text{Unif}[-1, 1]$ . Define

$$X = \sum_{n \geq 1} 2^{-n} U_n, \quad Z = 2X - 1 = \sum_{n \geq 1} 2^{-n} V_n. \quad (6)$$

Then

$$F(x) = \mathbb{P}(X \leq x), \quad \text{up}(x) \, dx = \mathbb{P}(Z \in dx). \quad (7)$$

Thus  $F$  is a CDF while  $\text{up}$  is the density of the centered and rescaled law. This classical probability representation goes back to the distribution-function viewpoint of Jessen–Wintner and Fabius [\[8, 5\]](#). It simultaneously explains positivity, monotonicity, symmetry, support, convolution formulas, and positive definiteness of the Fourier transform. The variance is

$$\mathbb{E}Z^2 = \frac{1}{3} \sum_{n \geq 1} 4^{-n} = \frac{1}{9}. \quad (8)$$

## 2.3 Fourier conventions

We use the angular transform

$$\varphi(t) = \int_{-1}^1 \text{up}(x) e^{-itx} dx = \mathbb{E} e^{-itZ}, \quad (9)$$

and the cycles-per-unit transform

$$\Phi(z) = \int_{-1}^1 \text{up}(x) e^{-2\pi izx} dx = \varphi(2\pi z). \quad (10)$$

We write  $\text{sinc } t = \sin t/t$  in angular formulas and explicitly write  $\sin(\pi z)/(\pi z)$  in normalized formulas.

## 3 A representation atlas from the audited corpus

This section is a compact, normalization-consistent synthesis of the principal representations already developed across the source tree. Later sections attach new operator meanings and consequences to this atlas.

### 3.1 Refinement and functional-differential forms

The bump satisfies the two-scale equation

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1), \quad (11)$$

with  $\text{up} \geq 0$ , support  $[-1, 1]$ , and integral one. Integrating or differentiating (11) produces a hierarchy of shifted/scaled formulas; folding by (2) recovers (1). On the signed extension, (5) removes the two shifts at the price of the Thue–Morse sign pattern.

A Volterra form useful for repeated integration is

$$F(x) = \int_0^{2x} F(t) dt \quad (x \geq 0), \quad (12)$$

where the globally signed continuation is understood once  $2x > 1$ . Iteration gives simplex integrals and triangular recurrences for dyadic samples.

### 3.2 Infinite convolution and finite box splines

Let

$$b_a(x) = \frac{1}{2a} \mathbf{1}_{[-a, a]}(x).$$

Then

$$\text{up} = b_{1/2} * b_{1/4} * b_{1/8} * \cdots \quad (13)$$

with convergence in every standard sense appropriate to the smooth limiting density. For the finite sum  $Z_N = \sum_{n=1}^N 2^{-n} V_n$ , write  $u_N$  for its density and  $a_N = 1 - 2^{-N}$ . Inclusion–exclusion gives the finite box-spline formula

$$u_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} (-1)^{s_2(j)} \left( x + a_N - \frac{j}{2^{N-1}} \right)_+^{N-1}. \quad (14)$$

This makes the Thue–Morse sequence visible before the smooth limit is taken. It is also a practical representation for derivatives of order below  $N$ , because each finite approximant is piecewise polynomial.

### 3.3 Dyadic sinc and cosine products

Independence in (6) yields

$$\varphi(t) = \prod_{n=1}^{\infty} \operatorname{sinc}\left(\frac{t}{2^n}\right), \quad \Phi(z) = \prod_{h=0}^{\infty} \frac{\sin(\pi z/2^h)}{\pi z/2^h}. \quad (15)$$

Using  $\operatorname{sinc} x = \prod_{j \geq 1} \cos(x/2^j)$  and counting pairs of scales gives

$$\varphi(t) = \prod_{r=1}^{\infty} \cos\left(\frac{t}{2^{r+1}}\right)^r, \quad \Phi(z) = \prod_{r=1}^{\infty} \cos\left(\frac{\pi z}{2^r}\right)^r. \quad (16)$$

The scale shift immediately gives the Mahler/refinement identities

$$\varphi(2t) = \operatorname{sinc}(t)\varphi(t), \quad \Phi(2z) = \frac{\sin(2\pi z)}{2\pi z}\Phi(z). \quad (17)$$

### 3.4 Collapsed canonical product and 2-adic multiplicity

Euler’s product for sine collapses all scales onto the positive integers:

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \quad (18)$$

Indeed, the factor at scale  $h$  vanishes at an integer  $m$  precisely when  $2^h \mid m$ , so

$$\operatorname{ord}_{z=\pm m} \Phi = 1 + \nu_2(m). \quad (19)$$

The associated Dirichlet series and exact counting function are

$$\sum_{m \geq 1} \frac{1 + \nu_2(m)}{m^s} = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1, \quad (20)$$

$$\sum_{m=1}^M (1 + \nu_2(m)) = 2M - s_2(M). \quad (21)$$

All of (18)–(21) are **corpus-known**; the new point in [section 4](#) is that the same multiplicity becomes the actual degeneracy of a specified self-adjoint operator.

### 3.5 Logarithmic, zeta, and Bernoulli series

Inside the first zero,  $|t| < 2\pi$ , logarithmic expansion of (18) gives

$$\log \varphi(t) = - \sum_{j \geq 1} \frac{t^{2j}}{j} \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (22)$$

Equivalently, for the moment generating function  $M_Z(s) = \varphi(is)$ ,

$$\log M_Z(s) = \sum_{j \geq 1} (-1)^{j+1} \frac{s^{2j}}{j} \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (23)$$

Therefore the even cumulants are

$$\kappa_{2j}(Z) = \frac{2^{2j-1} B_{2j}}{j(2^{2j} - 1)}, \quad \kappa_{2j+1}(Z) = 0. \quad (24)$$

Complete Bell polynomials then give

$$\mathbb{E}Z^{2n} = \mathcal{B}_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}). \quad (25)$$

The first values are

$$\mathbb{E}Z^2 = \frac{1}{9}, \quad \mathbb{E}Z^4 = \frac{19}{675}, \quad \mathbb{E}Z^6 = \frac{583}{59535}, \quad \mathbb{E}Z^8 = \frac{132809}{32531625}. \quad (26)$$

### 3.6 Fourier inversion, cosine series, and sampling

Because  $\text{up} \in C_c^\infty(\mathbb{R})$ , its transform is rapidly decreasing on the real axis, and Fourier inversion is absolutely convergent:

$$\text{up}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} \varphi(t) dt = \frac{1}{\pi} \int_0^{\infty} \cos(tx) \varphi(t) dt. \quad (27)$$

For  $0 < x < 1$ , the symmetric CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin((2x-1)t)}{t} \varphi(t) dt. \quad (28)$$

The period-two cosine expansion of the compact bump is

$$\text{up}(x) = \frac{1}{2} + \sum_{k \geq 0} \Phi\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi x), \quad |x| \leq 1. \quad (29)$$

Integer Fourier coefficients vanish because of (19), leaving only half-integers. Paley–Wiener sampling similarly yields

$$\Phi(z) = \frac{\sin(2\pi z)}{2\pi z} + \sum_{k \in \mathbb{Z}} \Phi\left(\frac{2k+1}{2}\right) \frac{\sin \pi(2z - 2k - 1)}{\pi(2z - 2k - 1)}, \quad (30)$$

with the standard  $L^2$  and locally uniform interpretations. Half-integer quadrature and arithmetic-ray refinements are developed extensively in the audited frontier reports.

### 3.7 Endpoint Laplace products

The moment generating function of  $X$  is

$$A(z) = \mathbb{E}e^{zX} = \prod_{n \geq 1} \frac{e^{z/2^n} - 1}{z/2^n} = e^{z/2} \varphi(iz/2). \quad (31)$$

The negative Laplace transform

$$L(s) = \mathbb{E}e^{-sX} = A(-s) = \prod_{n \geq 1} \frac{1 - e^{-s/2^n}}{s/2^n} = e^{-s/2} \varphi(is/2) \quad (32)$$

obeys

$$L(2s) = \frac{1 - e^{-s}}{s} L(s). \quad (33)$$



Writing  $\Lambda(s) = \log L(s)$ , the corpus derives the harmonic-sum Mellin transform

$$\int_0^\infty \Lambda(s) s^{w-1} ds = \frac{2^w}{1-2^w} [-\Gamma(w)\zeta(1+w)], \quad -1 < \Re w < 0, \quad (34)$$

and, after Mellin inversion, the logarithmic-periodic decomposition

$$\Lambda(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + P(\log_2 s) + E(s), \quad (35)$$

where  $P$  is nonconstant and one-periodic and  $E$  is rapidly small. This theory is **corpus-known**; [section 10](#) derives a Fredholm/Mellin–Barnes representation whose poles reproduce the same decomposition.

### 3.8 Exact reciprocal-dyadic values

Let  $d_n = \mathbb{E}X^n$ . A central exact identity in the corpus, closely related to the arithmetic formulas of Arias de Reyna [\[2\]](#), is

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{d_n}{n!} = 2^{-\binom{n}{2}} [z^n] A(z). \quad (36)$$

Among the many equivalent forms are the positive ordered-composition sum

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{(r_1, \dots, r_\ell) \models n} \prod_{j=1}^{\ell} \frac{1}{(2^{r_1 + \dots + r_j} - 1)(r_j + 1)!}, \quad (37)$$

Bell-polynomial and lower-Hessenberg determinant forms, and a finite half-base  $q$ -binomial formula of the shape

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2} (1/2; 1/2)_n}. \quad (38)$$

The numerator  $\mathcal{N}_n$  is an explicit alternating Gaussian-binomial sum with Thue–Morse inner sums. The new formula in [section 7](#) replaces these by a finite positive sum of exterior-power traces of one operator.

### 3.9 Endpoint and inverse asymptotics

For  $L_2 = \log 2$ , the lower-Lambert coordinate

$$\lambda_n = -\frac{1}{L_2} W_{-1}(-L_2 2^{-n}), \quad \lambda_n - \log_2 \lambda_n = n, \quad (39)$$

organizes the sharp reciprocal-power asymptotics. The audited frontier volume gives

$$\log F(2^{-n}) = -\frac{L_2}{2} \lambda_n^2 + \left(1 + \frac{L_2}{2}\right) \lambda_n - \frac{1}{2} \log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}), \quad (40)$$

where  $\Psi$  is smooth, nonconstant, mean-zero, and one-periodic, with explicit Gamma–zeta Fourier coefficients. All-order corrections use Bell polynomials. The same saddle machinery is inverted to describe  $F^{-1}(y)$  as  $y \downarrow 0$ . These results are **corpus-known** and provide the asymptotic benchmark for the new operator formulas below.

## 4 A specified Dirichlet operator and a genuine Fredholm determinant

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**Status.** The scalar sinc products, canonical product, zero multiplicities, and scalar spectral zeta in this section are **corpus-known**. The specified Hilbert space, Dirichlet Green operator, Fredholm determinant, unitary renormalization, and probabilistic operator consequences are **proved here** relative to the pinned corpus.

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## 4.1 The Green block

Let

$$D = -\frac{d^2}{dx^2}, \quad \mathcal{D}(D) = H^2(0, 1) \cap H_0^1(0, 1), \quad (41)$$

be the positive Dirichlet Laplacian on  $L^2(0, 1)$ . Its normalized eigenfunctions and eigenvalues are

$$e_k(x) = \sqrt{2} \sin(k\pi x), \quad De_k = \pi^2 k^2 e_k, \quad k \geq 1. \quad (42)$$

The inverse  $G = D^{-1}$  is the positive trace-class integral operator

$$(Gf)(x) = \int_0^1 g(x, y) f(y) dy, \quad g(x, y) = \min(x, y) - xy, \quad (43)$$

with

$$Ge_k = \frac{1}{\pi^2 k^2} e_k, \quad \text{Tr } G^s = \frac{\zeta(2s)}{\pi^{2s}} \quad (\Re s > \tfrac{1}{2}). \quad (44)$$

The same kernel is the covariance kernel of a standard Brownian bridge; that fact will be used in [section 6](#).

## 4.2 The dyadic direct sum

Set

$$\mathcal{H} = \bigoplus_{n=1}^{\infty} L^2(0, 1), \quad B = \bigoplus_{n=1}^{\infty} 4^{-n} G. \quad (45)$$

Since

$$\text{Tr } B = \sum_{n \geq 1} 4^{-n} \text{Tr } G = \frac{1}{3} \frac{\zeta(2)}{\pi^2} = \frac{1}{18}, \quad (46)$$

$B$  is positive trace class. Its nonzero eigenvalues, initially indexed by  $(n, k) \in \mathbb{N}^2$ , are

$$\lambda_{n,k} = \frac{4^{-n}}{\pi^2 k^2}. \quad (47)$$

**Theorem 4.1** (Specified Fredholm realization). *For every  $t \in \mathbb{C}$ ,*

$$\boxed{\varphi(t) = \det_{\mathcal{H}}(I - t^2 B).} \quad (48)$$

*Equivalently,*

$$\Phi(z) = \det_{\mathcal{H}}(I - 4\pi^2 z^2 B). \quad (49)$$

*Both are ordinary trace-class Fredholm determinants, not zeta-regularized products.*

*Proof.* The Fredholm determinant converges locally uniformly because  $B$  is trace class. Using (47) and Euler's sine product,

$$\begin{aligned} \det(I - t^2 B) &= \prod_{n \geq 1} \prod_{k \geq 1} \left( 1 - \frac{t^2}{\pi^2 4^n k^2} \right) \\ &= \prod_{n \geq 1} \frac{\sin(t/2^n)}{t/2^n} = \varphi(t). \end{aligned}$$

The normalized statement follows from  $\Phi(z) = \varphi(2\pi z)$ . □

This result gives an operator interpretation to every coefficient in the logarithmic series. For  $|t| < 2\pi$ ,

$$\log \varphi(t) = \operatorname{Tr} \log(I - t^2 B) = - \sum_{j \geq 1} \frac{t^{2j}}{j} \operatorname{Tr} B^j, \quad (50)$$

where

$$p_j := \operatorname{Tr} B^j = \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}. \quad (51)$$

Thus (22) is literally a trace expansion.

### 4.3 A self-adjoint operator whose degeneracies are 2-adic

Let

$$\mathcal{L} = B^{-1} = \bigoplus_{n \geq 1} 4^n D \quad (52)$$

with its natural direct-sum domain. This is positive self-adjoint with compact resolvent. Reindexing  $m = 2^{n-1}k$  in (47) gives

$$\operatorname{Spec}(\mathcal{L}) = \{(2\pi m)^2 : m \geq 1\}, \quad \operatorname{mult}_{(2\pi m)^2} \mathcal{L} = 1 + \nu_2(m). \quad (53)$$

Indeed,  $m = 2^{n-1}k$  is possible for exactly  $n = 1, \dots, 1 + \nu_2(m)$ . Therefore

$$\varphi(t) = \det(I - t^2 \mathcal{L}^{-1}) = \prod_{m \geq 1} \left(1 - \frac{t^2}{(2\pi m)^2}\right)^{1 + \nu_2(m)}. \quad (54)$$

This proves that the 2-adic zero multiplicity is an actual spectral degeneracy, not merely a multiplicity in a formal zero divisor.

The spectral zeta function of  $\mathcal{L}$  is

$$\zeta_{\mathcal{L}}(s) = \operatorname{Tr} \mathcal{L}^{-s} = \frac{\zeta(2s)}{\pi^{2s}(4^s - 1)}, \quad \Re s > \frac{1}{2}, \quad (55)$$

which extends meromorphically through the right-hand side. Its poles at  $s = \pi i k / \log 2$ , including the collision at  $s = 0$ , are the operator source of the logarithmic-periodic oscillations encountered in endpoint asymptotics.

### 4.4 Exact renormalization by removing one block

The dyadic direct sum satisfies the unitary equivalences

$$4B \simeq G \oplus B, \quad \mathcal{L} \simeq 4D \oplus 4\mathcal{L}. \quad (56)$$

Taking determinants in the first identity gives

$$\begin{aligned} \det(I - 4t^2 B) &= \det(I - t^2 G) \det(I - t^2 B), \\ \varphi(2t) &= \operatorname{sinc}(t) \varphi(t), \end{aligned} \quad (57)$$

so the familiar Fourier scaling law is an exact block-decimation identity. On the imaginary axis, if

$$D_B(q) = \det(I + qB), \quad q \in \mathbb{C}, \quad (58)$$

then

$$D_B(4q) = \frac{\sinh \sqrt{q}}{\sqrt{q}} D_B(q). \quad (59)$$

Equations (33) and (59) are the same renormalization after

$$L(s) = e^{-s/2} D_B(s^2/4). \quad (60)$$

## 4.5 Geometric-scale and exponent-sequence operators

The construction is not confined to the classical scale. Let  $0 < q < 1$ ,  $c > 0$ , and let  $a_n \in \mathbb{Z}_{\geq 0}$  satisfy

$$\sum_{n \geq 0} a_n c^2 q^{2n} < \infty. \quad (61)$$

On a direct sum with  $a_n$  copies of  $L^2(0, 1)$  at scale  $n$ , put

$$B_{a,q,c} = \bigoplus_{n \geq 0} \left( c^2 q^{2n} G \right)^{\oplus a_n}. \quad (62)$$

Then

$$\det(I - t^2 B_{a,q,c}) = \prod_{n \geq 0} \operatorname{sinc}(cq^n t)^{a_n}, \quad (63)$$

while

$$\operatorname{Tr} B_{a,q,c}^s = \frac{\zeta(2s)}{\pi^{2s}} c^{2s} \sum_{n \geq 0} a_n q^{2ns}. \quad (64)$$

For  $a_n \equiv 1$ , this becomes

$$\operatorname{Tr} B_{q,c}^s = \frac{c^{2s} \zeta(2s)}{\pi^{2s} (1 - q^{2s})}. \quad (65)$$

The classical Fabius–Rvachev operator is  $q = c = 1/2$ . Thus the exponent-sequence Dirichlet series in the audited hierarchy is realized as an honest operator trace. In particular, scalar “zero zeta” identities become spectral zeta identities whenever the exponents are nonnegative integers.

## 5 Resolvent, heat, theta, and Stieltjes representations

### 5.1 The logarithmic derivative as a resolvent trace

For  $t$  away from the zeros of  $\varphi$ , differentiating the Fredholm determinant gives

$$\frac{\varphi'(t)}{\varphi(t)} = -2t \operatorname{Tr}[B(I - t^2 B)^{-1}]. \quad (66)$$

Using either spectral indexing yields two complementary partial-fraction forms:

$$\frac{\varphi'(t)}{\varphi(t)} = -2t \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 - t^2}, \quad (67)$$

$$= \sum_{n \geq 1} \left[ 2^{-n} \cot\left(\frac{t}{2^n}\right) - \frac{1}{t} \right]. \quad (68)$$

Although each bracket in (68) contains  $-1/t$ , the bracket as a whole is  $\mathcal{O}(4^{-n})$  on compact sets away from poles, so the series is locally normally convergent. Equating (67) and (68) gives a 2-adic cotangent identity:

$$\sum_{n \geq 1} \left[ 2^{-n} \cot\left(\frac{t}{2^n}\right) - \frac{1}{t} \right] = -2t \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 - t^2}. \quad (69)$$

This is simultaneously a scale expansion and a spectral resolvent expansion.

For the positive determinant  $D_B(q)$ ,

$$\frac{d}{dq} \log D_B(q) = \operatorname{Tr}[B(I + qB)^{-1}] = \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 + q}. \quad (70)$$

Thus the logarithmic derivative is the Stieltjes transform of the discrete spectral measure

$$\rho = \sum_{m \geq 1} (1 + \nu_2(m)) \delta_{(2\pi m)^2}. \quad (71)$$

## 5.2 Heat trace and dyadic theta series

Define

$$H(x) = \operatorname{Tr} e^{-x\mathcal{L}} = \sum_{n \geq 1} \sum_{k \geq 1} e^{-\pi^2 4^n k^2 x} = \sum_{m \geq 1} (1 + \nu_2(m)) e^{-(2\pi m)^2 x}. \quad (72)$$

In terms of Jacobi's theta function  $\vartheta_3(0, q) = 1 + 2 \sum_{k \geq 1} q^{k^2}$ ,

$$H(x) = \frac{1}{2} \sum_{n \geq 1} [\vartheta_3(0, e^{-\pi^2 4^n x}) - 1]. \quad (73)$$

Removing the first block gives the exact heat renormalization

$$H(x/4) = \sum_{k \geq 1} e^{-\pi^2 k^2 x} + H(x). \quad (74)$$

The Mellin transform is

$$\int_0^\infty x^{s-1} H(x) dx = \Gamma(s) \frac{\zeta(2s)}{\pi^{2s} (4^s - 1)}, \quad \Re s > \frac{1}{2}. \quad (75)$$

This is a theta-integral representation of the spectral zeta and a direct route from Poisson summation to the endpoint logarithmic period.

## 5.3 Exponentials of heat integrals

The elementary identity

$$\log(1 + q\lambda) = \int_0^\infty \frac{1 - e^{-qx}}{x} e^{-x/\lambda} dx, \quad q \geq 0, \quad (76)$$

can be summed over the spectrum because  $B$  is trace class.

**Theorem 5.1** (Heat-kernel determinant representations). *For  $q \geq 0$ ,*

$$\log D_B(q) = \int_0^\infty \frac{1 - e^{-qx}}{x} H(x) dx. \quad (77)$$

For  $|t| < 2\pi$ ,

$$\log \varphi(t) = - \int_0^\infty \frac{e^{t^2 x} - 1}{x} H(x) dx. \quad (78)$$

The latter identity extends by analytic continuation from the disk bounded by the first zero.

*Proof.* Apply (76) to each eigenvalue  $\lambda_j$  and use monotone convergence for  $q \geq 0$ . Since  $\sum_j e^{-x/\lambda_j} = H(x)$ , this proves (77). Replacing  $q$  by  $-t^2$  is justified when  $t^2 < \lambda_1^{-1} = 4\pi^2$ , and gives (78).  $\square$

Combining (77) and (73) produces the explicit product–integral–theta representation

$$\varphi(i\sqrt{q}) = \exp \left\{ \frac{1}{2} \sum_{n \geq 1} \int_0^\infty \frac{1 - e^{-qx}}{x} [\vartheta_3(0, e^{-\pi^2 4^n x}) - 1] dx \right\}. \quad (79)$$

## 5.4 Complete Bernstein structure

Let  $f(q) = \log D_B(q)$  for  $q \geq 0$ . From (70),

$$f'(q) = \int_{(0,\infty)} \frac{\rho(dy)}{q+y} \quad (80)$$

with  $\rho$  from (71). Hence  $f'$  is a Stieltjes function, and  $f$  is a complete Bernstein function in the sense of [14]. Equivalently, its Lévy–Khintchine representation is exactly (77), whose Lévy density is

$$\frac{H(x)}{x} = \frac{1}{x} \sum_{m \geq 1} (1 + \nu_2(m)) e^{-(2\pi m)^2 x}. \quad (81)$$

In particular,

$$(-1)^r f^{(r+1)}(q) = r! \sum_j \frac{\lambda_j^{r+1}}{(1 + q\lambda_j)^{r+1}} > 0, \quad r \geq 0. \quad (82)$$

This supplies an infinite family of monotonicity and convexity inequalities for the imaginary-axis Fourier transform.

## 6 Probability laws generated by the operator

The Dirichlet block has an immediate probabilistic meaning that is not visible in the scalar canonical product. This section turns the determinant and its reciprocal powers into Laplace transforms of explicit random series.

### 6.1 Brownian-bridge quadratic functionals

Let  $\beta$  be a standard Brownian bridge on  $[0, 1]$ . Its Karhunen–Loève expansion is [9]

$$\beta(x) = \sum_{k \geq 1} \frac{\xi_k}{\pi k} \sqrt{2} \sin(\pi k x), \quad \xi_k \stackrel{\text{iid}}{\sim} N(0, 1), \quad (83)$$

and hence

$$\int_0^1 \beta(x)^2 dx = \sum_{k \geq 1} \frac{\xi_k^2}{\pi^2 k^2} \quad \text{almost surely and in } L^1. \quad (84)$$

Take independent bridges  $\beta_n$ ,  $n \geq 1$ , and define

$$Q = \sum_{n \geq 1} 4^{-n} \int_0^1 \beta_n(x)^2 dx. \quad (85)$$

The series converges almost surely because  $\mathbb{E}Q = \text{Tr } B = 1/18$ .

**Theorem 6.1** (Brownian-bridge determinant law). *For every  $q \geq 0$ ,*

$$\mathbb{E} \exp\left(-\frac{q}{2} Q\right) = D_B(q)^{-1/2} = \varphi(i\sqrt{q})^{-1/2}. \quad (86)$$

*Equivalently, after grouping equal eigenvalues,*

$$\frac{Q}{2} \stackrel{d}{=} \sum_{m \geq 1} \Gamma_m, \quad \Gamma_m \sim \text{Gamma}\left(\frac{1 + \nu_2(m)}{2}, \frac{1}{(2\pi m)^2}\right), \quad (87)$$

*where the second parameter is the scale and the variables are independent.*

*Proof.* Insert (84) into (85). The coefficients of the independent squares  $\xi_{n,k}^2$  are exactly the eigenvalues  $\lambda_{n,k}$  of  $B$ . For a standard normal variable,  $\mathbb{E} \exp(-q\lambda\xi^2/2) = (1 + q\lambda)^{-1/2}$ . Multiplication over the spectrum proves (86). Reindexing by  $m = 2^{n-1}k$  collects  $1 + \nu_2(m)$  chi-square degrees of freedom at scale  $(2\pi m)^{-2}$ , which is (87).  $\square$

This result places the Rvachev Fourier image beside classical quadratic-functionals such as the Cramér–von Mises statistic, but with an infinite dyadic tower of independent bridges. In particular, every fractional reciprocal power  $D_B(q)^{-\alpha}$ ,  $\alpha > 0$ , is completely monotone: replace each gamma shape in (87) by  $\alpha(1 + \nu_2(m))$ .

## 6.2 An exponential series and a generalized gamma convolution

Let  $E_{n,k}$  be independent unit exponential random variables and set

$$S = \sum_{n,k \geq 1} \lambda_{n,k} E_{n,k}. \quad (88)$$

Again  $\mathbb{E}S = \text{Tr } B < \infty$ , so the series converges almost surely.

**Proposition 6.2** (Reciprocal determinant as a Laplace transform). *For  $q \geq 0$ ,*

$$\boxed{\mathbb{E}e^{-qS} = D_B(q)^{-1}.} \quad (89)$$

Moreover,

$$S \stackrel{d}{=} \sum_{m \geq 1} Y_m, \quad Y_m \sim \text{Gamma}\left(1 + \nu_2(m), \frac{1}{(2\pi m)^2}\right) \quad (90)$$

independently.

The cumulants of  $S$  are therefore

$$\kappa_j(S) = (j-1)! \text{Tr } B^j = (j-1)! \frac{\zeta(2j)}{\pi^{2j}(4^j - 1)}, \quad j \geq 1. \quad (91)$$

If  $h_j(B)$  denotes the complete homogeneous symmetric function of degree  $j$  in the eigenvalues of  $B$ , then

$$\frac{1}{D_B(-q)} = \sum_{j \geq 0} h_j(B) q^j, \quad \mathbb{E}S^j = j! h_j(B). \quad (92)$$

Thus the determinant simultaneously packages two dual symmetric-function families: its coefficients are exterior powers, while the reciprocal coefficients are symmetric powers.

## 6.3 A Thorin and heat-measure description

Equation (89) shows that  $S$  is a generalized gamma convolution with discrete Thorin measure

$$U = \sum_{m \geq 1} (1 + \nu_2(m)) \delta_{(2\pi m)^2}. \quad (93)$$

Indeed,

$$-\log \mathbb{E}e^{-qS} = \int_{(0,\infty)} \log\left(1 + \frac{q}{y}\right) U(dy), \quad (94)$$

which is the same formula as (77). The Lévy density  $H(x)/x$  is therefore not an auxiliary analytic device: it is the Lévy density of an explicit infinitely divisible probability law. This creates a three-way dictionary

dyadic sinc product  $\longleftrightarrow$  Dirichlet spectrum  $\longleftrightarrow$  generalized gamma convolution.

## 7 Exterior powers, exact moments, and reciprocal-dyadic values

### 7.1 Exterior powers are normalized even moments

For a positive trace-class operator, the Fredholm determinant has the absolutely convergent trace-ideal expansion [16, 7]

$$D_B(q) = \det(I + qB) = \sum_{k=0}^{\infty} e_k(B) q^k, \quad e_k(B) = \text{Tr}(\wedge^k B). \quad (95)$$

On the other hand, by the probability model,

$$D_B(s^2) = \varphi(is) = \mathbb{E} e^{sZ} = \sum_{k=0}^{\infty} \frac{\mathbb{E} Z^{2k}}{(2k)!} s^{2k}. \quad (96)$$

Comparison proves the following exact identification.

**Theorem 7.1** (Moment–exterior-power identity). *For every  $k \geq 0$ ,*

$$e_k(B) = \text{Tr}(\wedge^k B) = \frac{\mathbb{E} Z^{2k}}{(2k)!}. \quad (97)$$

*In particular all exterior-power traces are positive rational numbers.*

The rationality follows from the power traces  $p_j = \text{Tr} B^j$ , since Euler’s evaluation of  $\zeta(2j)$  gives

$$p_j = \frac{(-1)^{j+1} 2^{2j-1} B_{2j}}{(2j)!(4j-1)} \in \mathbb{Q}_{>0}. \quad (98)$$

Newton’s identities become

$$n e_n = \sum_{j=1}^n (-1)^{j-1} p_j e_{n-j}, \quad e_0 = 1, \quad (99)$$

and the determinant form is

$$e_n = \frac{1}{n!} \det \begin{pmatrix} p_1 & 1 & 0 & \cdots & 0 \\ p_2 & p_1 & 2 & \cdots & 0 \\ p_3 & p_2 & p_1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & n-1 \\ p_n & p_{n-1} & \cdots & p_2 & p_1 \end{pmatrix}. \quad (100)$$

Equivalently, the partition expansion is

$$e_n = \sum_{m_1+2m_2+\cdots+nm_n=n} \prod_{j=1}^n \frac{1}{m_j!} \left( \frac{(-1)^{j-1} p_j}{j} \right)^{m_j}. \quad (101)$$

Although signs occur in (101), positivity is transparent from the spectral sum

$$e_n = \sum_{j_1 < \cdots < j_n} \lambda_{j_1} \cdots \lambda_{j_n}. \quad (102)$$

The first values are

$$e_0 = 1, \quad e_1 = \frac{1}{18}, \quad e_2 = \frac{19}{16200}, \quad e_3 = \frac{583}{42865200}, \quad e_4 = \frac{132809}{1311675120000}. \quad (103)$$



## 7.2 A positive spectral formula for $F(2^{-n})$

The audited corpus supplies (36). The centered variable gives

$$\mathbb{E}X^n = 2^{-n}\mathbb{E}(1+Z)^n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} \mathbb{E}Z^{2k}. \quad (104)$$

Substituting (97) into (36) proves the promised exact representation.

**Theorem 7.2** (Exterior-power formula for reciprocal dyadic values). *For every integer  $n \geq 0$ ,*

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\text{Tr}(\wedge^k B)}{(n-2k)!}. \quad (105)$$

Equivalently,

$$2^{n(n+1)/2}F(2^{-n}) = [z^n]e^z D_B(z^2). \quad (106)$$

The formula is a finite positive sum even though each exterior trace is itself an infinite multiple series. Combining (105) with (101) gives a finite Bernoulli–partition expression:

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{(n-2k)!} \sum_{m_1+2m_2+\dots+km_k=k} \prod_{j=1}^k \frac{1}{m_j!} \left[ \frac{(-1)^{j-1}}{j} \frac{\zeta(2j)}{\pi^{2j}(4^j-1)} \right]^{m_j}. \quad (107)$$

A purely positive multiple-zeta form follows from (102):

$$F(2^{-n}) = 2^{-n(n+1)/2} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{1}{(n-2k)!} \sum_{j_1 < \dots < j_k} \prod_{r=1}^k \lambda_{j_r}. \quad (108)$$

Here the single index  $j$  may be replaced by either the pair  $(a, b)$  with  $\lambda_{a,b} = 4^{-a}/(\pi^2 b^2)$ , or the collapsed integer index with multiplicity  $1 + \nu_2(m)$ .

Table 1: Exact values produced by (105).

| $n$ | $2^{-n}$ | $F(2^{-n})$                   |
|-----|----------|-------------------------------|
| 0   | 1        | 1                             |
| 1   | 1/2      | 1/2                           |
| 2   | 1/4      | 5/72                          |
| 3   | 1/8      | 1/288                         |
| 4   | 1/16     | 143/2073600                   |
| 5   | 1/32     | 19/33177600                   |
| 6   | 1/64     | 1153/561842749440             |
| 7   | 1/128    | 583/179789679820800           |
| 8   | 1/256    | 1616353/704200217922109440000 |

## 7.3 A contour coefficient representation

Cauchy’s formula applied to (106) yields, for any positively oriented circle  $\gamma$  around the origin,

$$F(2^{-n}) = \frac{2^{-n(n+1)/2}}{2\pi i} \int_{\gamma} \frac{e^z \det(I + z^2 B)}{z^{n+1}} dz. \quad (109)$$

This representation invites saddle-point analysis directly on the Fredholm determinant. Its stationary equation is the resolvent equation

$$1 + 2z \operatorname{Tr}[B(I + z^2 B)^{-1}] - \frac{n+1}{z} = 0, \quad (110)$$

which is an operator counterpart of the lower-Lambert saddle used in the corpus.

## 8 Pólya-frequency structure and real-zero consequences

### 8.1 Toeplitz total positivity

The product

$$D_B(q) = \prod_{j \geq 1} (1 + \lambda_j q), \quad \lambda_j > 0, \quad \sum_j \lambda_j < \infty, \quad (111)$$

places its coefficient sequence in the classical Aissen–Schoenberg–Whitney–Edrei class [3, 4, 15].

**Theorem 8.1** (Pólya-frequency theorem for normalized even moments). *The sequence*

$$a_n = e_n(B) = \frac{\mathbb{E} Z^{2n}}{(2n)!}, \quad n \geq 0, \quad (112)$$

*is a Pólya-frequency sequence of infinite order. Equivalently, every finite minor of the Toeplitz matrix*

$$T(a) = (a_{j-i})_{i,j \geq 0}, \quad a_r = 0 \ (r < 0), \quad (113)$$

*is nonnegative.*

*Proof.* Every finite product  $\prod_{j=1}^N (1 + \lambda_j q)$  has only real nonpositive zeros, so its coefficient sequence is  $\text{PF}_\infty$ . Since  $\sum_j \lambda_j < \infty$ , the finite products converge locally uniformly to  $D_B$ , and every fixed Toeplitz minor converges to the corresponding minor of  $T(a)$ . Total positivity is closed under this limit.  $\square$

The dual Jacobi–Trudi identity gives more information. If  $\lambda$  and  $\mu$  are partitions with  $\mu \subseteq \lambda$ , then an appropriate Toeplitz minor is the skew Schur function

$$\det(e_{\lambda'_i - \mu'_j - i + j}(B)) = s_{\lambda/\mu}(\lambda_1, \lambda_2, \dots) \geq 0. \quad (114)$$

Through the Frobenius character formula, every such minor is an explicit finite combination of the power traces (51), and hence of Bernoulli numbers. This extends the single sequence of moments to an infinite hierarchy of spectral inequalities.

### 8.2 Strict Newton–Turán inequalities

Newton’s inequalities for entire functions of Laguerre–Pólya type imply strong log-concavity of the factorially normalized coefficients.

**Corollary 8.2** (Strict Newton inequalities). *For every  $n \geq 1$ ,*

$$e_n(B)^2 > \frac{n+1}{n} e_{n-1}(B) e_{n+1}(B). \quad (115)$$

*Equivalently, the moment sequence satisfies*

$$\left( \frac{\mathbb{E} Z^{2n}}{(2n)!} \right)^2 > \frac{n+1}{n} \frac{\mathbb{E} Z^{2n-2}}{(2n-2)!} \frac{\mathbb{E} Z^{2n+2}}{(2n+2)!}. \quad (116)$$

Equivalently, after normalizing by  $D_B(1)$ , the coefficients are the law of a countable Bernoulli sum and are ultra-log-concave [11]. Strictness follows because  $D_B$  has infinitely many nonzero roots; the equality case of the limiting Newton inequality is the zero-free exponential  $e^{cq}$ , not the determinant (111). Numerically,  $e_1^2/(e_0 e_2) = 50/19 \approx 2.63158$ , while the lower bound is 2.

### 8.3 Laguerre–Pólya class and derivative zeros

The finite determinant truncations are polynomials with real zeros  $\pm\lambda_j^{-1/2}$ . They converge locally uniformly to  $\varphi$ , so

$$\varphi \in \mathcal{LP}, \quad (117)$$

where  $\mathcal{LP}$  is the Laguerre–Pólya class. Therefore every derivative  $\varphi^{(r)}$  has only real zeros. At  $t = 2\pi m$ , differentiation removes one unit of the known multiplicity until it is exhausted; all remaining derivative zeros are constrained to the real gaps between the spectral points. This elementary but strong consequence is useful for validating numerical Fourier calculations and motivates [theorem 12.1](#) below.

The power traces also give Euler–Rayleigh bounds for the first positive zero  $2\pi$ : for every  $j \geq 1$ ,

$$p_j^{-1/(2j)} < 2\pi < \sqrt{\frac{p_j}{p_{j+1}}}, \quad p_j = \frac{\zeta(2j)}{\pi^{2j}(4j-1)}. \quad (118)$$

Although the zero is already known exactly, these inequalities illustrate how the Bernoulli trace data alone recover the spectral edge.

## 9 Determinant inversion formulas for up and $F$

Substitution of the Fredholm determinant into classical inversion formulas produces representations that involve no unnamed Fourier transform.

### 9.1 Real Fourier integrals

For every  $x \in \mathbb{R}$ ,

$$\text{up}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} \det(I - t^2 B) dt = \frac{1}{\pi} \int_0^{\infty} \cos(tx) \det(I - t^2 B) dt. \quad (119)$$

For  $0 < x < 1$ ,

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \frac{\sin((2x-1)t)}{t} \det(I - t^2 B) dt. \quad (120)$$

The integrands are Schwartz on the real axis, despite the determinant's exponential type in nonreal directions. Differentiating (120) recovers  $F'(x) = 2 \text{up}(2x-1)$ .

### 9.2 Bromwich–Fredholm inversion

Since  $L(s) = \int_{[0,1]} e^{-sx} dF(x)$ , integration by parts gives

$$\int_0^1 e^{-sx} F(x) dx = \frac{L(s) - e^{-s}}{s}. \quad (121)$$

It is often cleaner to invert the measure directly. At every  $0 < x < 1$  and any  $c > 0$ , the standard Laplace–Stieltjes inversion formula gives

$$F(x) = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{e^{sx}}{s} L(s) ds. \quad (122)$$

Using (60),

$$F(x) = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{e^{s(x-1/2)}}{s} \det\left(I + \frac{s^2}{4} B\right) ds. \quad (123)$$

Similarly, at interior points,

$$2 \operatorname{up}(2x - 1) = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} e^{s(x-1/2)} \det \left( I + \frac{s^2}{4} B \right) ds. \quad (124)$$

These contour formulas are particularly suited to saddle deformation, because their phase derivative is the explicit resolvent trace (70).

### 9.3 A determinant series on compact subdisks

Within  $|t| < 2\pi$ , exponentiating the trace series gives

$$\det(I - t^2 B) = \exp \left[ - \sum_{j \geq 1} \frac{p_j}{j} t^{2j} \right] = \sum_{n \geq 0} (-1)^n e_n(B) t^{2n}. \quad (125)$$

Substituting the first exponential in (120) is useful after splitting the Fourier integral at any radius below  $2\pi$ ; substituting the second gives a moment/exterior-power expansion of the low-frequency contribution. This is not a global termwise Fourier inversion—the Taylor series is local—but it provides rigorous error-controlled hybrid quadrature when paired with the superpolynomial decay estimates from the audited Fourier dossiers.

## 10 A Mellin–Barnes Fredholm representation of the endpoint transform

The corpus’s Mellin harmonic-sum analysis and the new determinant model meet in a particularly direct contour integral. The bridge is the classical Mellin–Barnes identity [6]:

$$\log(1 + x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{u \sin \pi u} x^u du, \quad x > 0, \quad 0 < c < 1. \quad (126)$$

**Theorem 10.1** (Endpoint Mellin–Barnes determinant formula). *Let  $s > 0$  and  $1/2 < c < 1$ . Then*

$$\log L(s) = -\frac{s}{2} + \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{u \sin \pi u} \left( \frac{s}{2\pi} \right)^{2u} \frac{\zeta(2u)}{4^u - 1} du. \quad (127)$$

*The integral is absolutely convergent on the vertical line.*

*Proof.* By (60),

$$\log L(s) = -\frac{s}{2} + \sum_j \log \left( 1 + \frac{s^2}{4} \lambda_j \right).$$

Apply (126) term by term. The choice  $c > 1/2$  ensures  $\sum_j \lambda_j^c = \operatorname{Tr} B^c < \infty$ , so Fubini’s theorem is applicable. The spectral trace (55) then gives

$$\left( \frac{s^2}{4} \right)^u \operatorname{Tr} B^u = \left( \frac{s}{2\pi} \right)^{2u} \frac{\zeta(2u)}{4^u - 1}.$$

The factor  $1/\sin \pi u$  supplies exponential decay on vertical lines, while the remaining factors have at most polynomial growth.  $\square$

## 10.1 Pole geometry and the periodic endpoint term

The integrand in (127) has three kinds of relevant singularities.

(i) At  $u = 1/2$ , the pole of  $\zeta(2u)$  has residue  $s/2$ , exactly cancelling the explicit term  $-s/2$ .

(ii) At

$$\tau_k = \frac{\pi i k}{\log 2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad (128)$$

the zeros of  $4^u - 1$  produce the logarithmic-periodic Fourier modes.

(iii) At  $u = 0$ , the collision of the pole of  $1/(4^u - 1)$  with the double pole of  $\pi/(u \sin \pi u)$  produces a triple pole and hence the quadratic polynomial in  $\log s$ .

Put  $L_2 = \log 2$ ,  $a = \log(2\pi)$ , and

$$C_0 = \frac{a^2/2 + \zeta''(0) - \pi^2/24}{L_2} - \frac{L_2}{12}. \quad (129)$$

A Laurent expansion at zero gives the residue

$$-\frac{(\log s)^2}{2L_2} + \frac{1}{2} \log s + C_0. \quad (130)$$

For  $k \neq 0$ , define

$$c_k = \frac{\pi \zeta(2\tau_k)}{2L_2 \tau_k \sin(\pi \tau_k)}. \quad (131)$$

Then  $c_{-k} = \overline{c_k}$ , and the exponential size of  $\sin(\pi \tau_k)$  implies rapid convergence of the real periodic function

$$P_{\text{MB}}(y) = \sum_{k \neq 0} c_k e^{2\pi i k y}. \quad (132)$$

Shifting the contour across the imaginary-axis poles yields, for every  $A > 0$ ,

$$\log L(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + C_0 + P_{\text{MB}}\left(\log_2 \frac{s}{2\pi}\right) + \mathcal{O}_A(s^{-A}). \quad (133)$$

The negative integers cause no additional residues: the poles of  $1/\sin \pi u$  are cancelled by the trivial zeros  $\zeta(2u) = 0$ . Formula (133) is the Fredholm-residue version of the corpus-known endpoint periodicity. Standard functional-equation manipulations convert (131) to the Gamma-zeta coefficients used in the harmonic-sum derivation.

**Remark 10.2** (What the contour shift proves). The contour may be moved to an arbitrary fixed left half-plane, giving a remainder smaller than every fixed negative power of  $s$ . The numerical data and the original dyadic product indicate a much sharper exponentially small remainder. A complete uniform transseries statement is isolated as [theorem 12.5](#) rather than being assumed here.

## 10.2 A second Mellin representation through the heat trace

Combining (77) with (60) gives

$$\log L(s) = -\frac{s}{2} + \int_0^\infty \frac{1 - e^{-s^2 x/4}}{x} H(x) dx. \quad (134)$$

Inserting (73) produces the mixed theta–Laplace integral

$$\log L(s) = -\frac{s}{2} + \frac{1}{2} \sum_{n \geq 1} \int_0^\infty \frac{1 - e^{-s^2 x/4}}{x} \left[ \vartheta_3(0, e^{-\pi^2 4^n x}) - 1 \right] dx. \quad (135)$$

Taking the Mellin transform in  $s^2/4$  recovers (127). Thus the endpoint oscillation can be viewed in three equivalent ways: poles of the scale-harmonic Mellin transform, complex dimensions of the Dirichlet tower, or complex poles of the Fredholm spectral zeta.

## 11 Numerical experiments and reproducibility

The archive contains `numerical_experiments.py`, written with detailed comments and using `mpmath`, `sympy`, and `matplotlib`. It regenerates every CSV table and figure in this section. All transcendental checks use 50 decimal digits; the exact moment and dyadic calculations use rational arithmetic.

### 11.1 Fourier product versus trace exponential

For  $|t| < 2\pi$ , the code compares

$$\prod_{n=1}^{120} \operatorname{sinc}(t/2^n) \quad \text{with} \quad \exp\left(-\sum_{j=1}^{180} \frac{p_j t^{2j}}{j}\right).$$

Representative values are shown in [table 2](#).

Table 2: Dyadic product and trace-series agreement.

| $t$  | dyadic sinc product | trace exponential  | absolute error        |
|------|---------------------|--------------------|-----------------------|
| 0.25 | 0.9965323558631412  | 0.9965323558631412 | $1.6 \times 10^{-50}$ |
| 0.75 | 0.9691186832108649  | 0.9691186832108649 | $1.6 \times 10^{-50}$ |
| 1.50 | 0.8807851433857338  | 0.8807851433857338 | $9.4 \times 10^{-51}$ |
| 2.75 | 0.6413734517278066  | 0.6413734517278066 | $2.3 \times 10^{-50}$ |
| 4.00 | 0.3617631389237471  | 0.3617631389237471 | $2.7 \times 10^{-51}$ |
| 5.25 | 0.1258123621455857  | 0.1258123621455857 | $1.3 \times 10^{-31}$ |

The slower convergence near  $t = 5.25$  reflects the radius  $2\pi$  set by the first spectral zero.

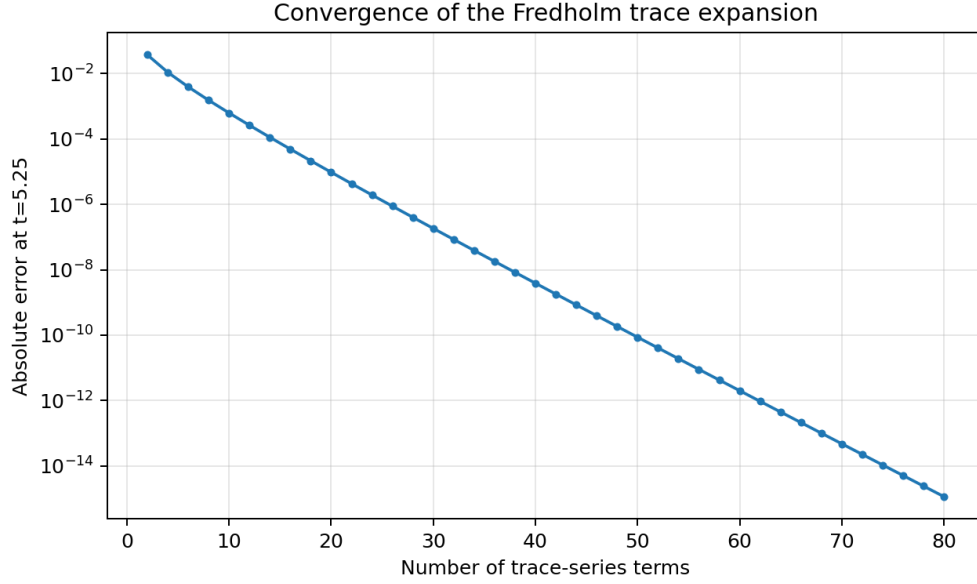


Figure 1: Convergence of the trace expansion at  $t = 5.25$ .

## 11.2 Exact exterior powers and Turán ratios

Newton's identities were evaluated exactly through  $e_{18}$ , and (105) was checked through  $n = 15$ . The generated files `exterior_coefficients.csv` and `dyadic_exterior_values.csv` contain all fractions. The strict inequality ratio

$$R_n = \frac{e_n^2}{e_{n-1}e_{n+1}} \quad (136)$$

was compared with  $(n+1)/n$  for  $1 \leq n \leq 14$ . The gaps remain positive and decrease smoothly, as shown in figure 2.

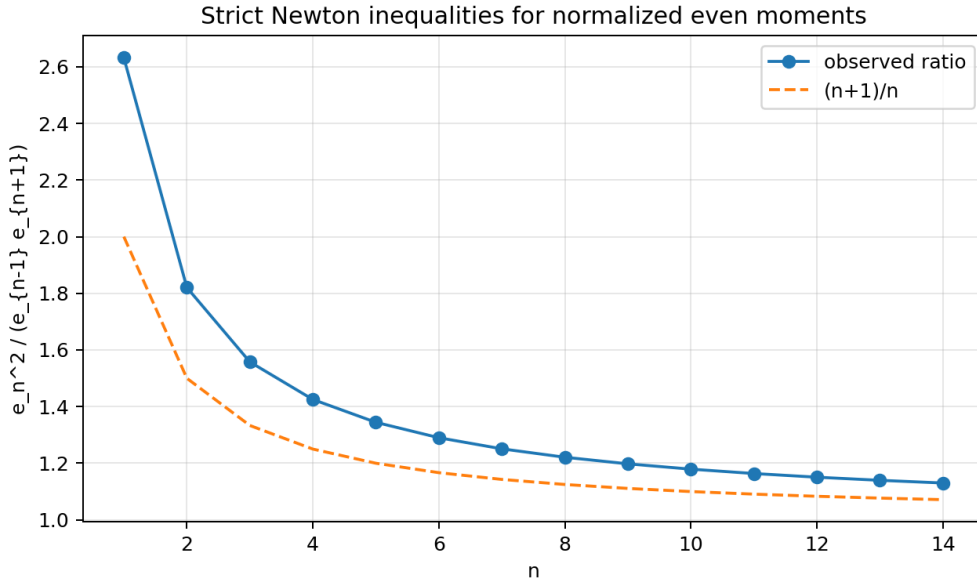


Figure 2: Observed Newton–Turán ratios and the sharp limiting lower bound.

### 11.3 Mellin–Barnes validation

The vertical integral in (127) was computed on  $\Re u = 3/4$ , using conjugation symmetry and a cutoff  $|\Im u| = 35$ . Direct evaluation of the endpoint product and the contour integral agree as follows.

Table 3: Numerical check of the Mellin–Barnes formula.

| $s$ | $\log L(s)$ , product | $\log L(s)$ , MB    | absolute error        |
|-----|-----------------------|---------------------|-----------------------|
| 0.5 | -0.2465292231716876   | -0.2465292231716876 | $1.0 \times 10^{-50}$ |
| 2   | -0.9448093175660127   | -0.9448093175660127 | $1.2 \times 10^{-50}$ |
| 10  | -3.782217787719464    | -3.782217787719464  | $3.2 \times 10^{-50}$ |
| 50  | -10.19248359859169    | -10.19248359859169  | $1.7 \times 10^{-49}$ |

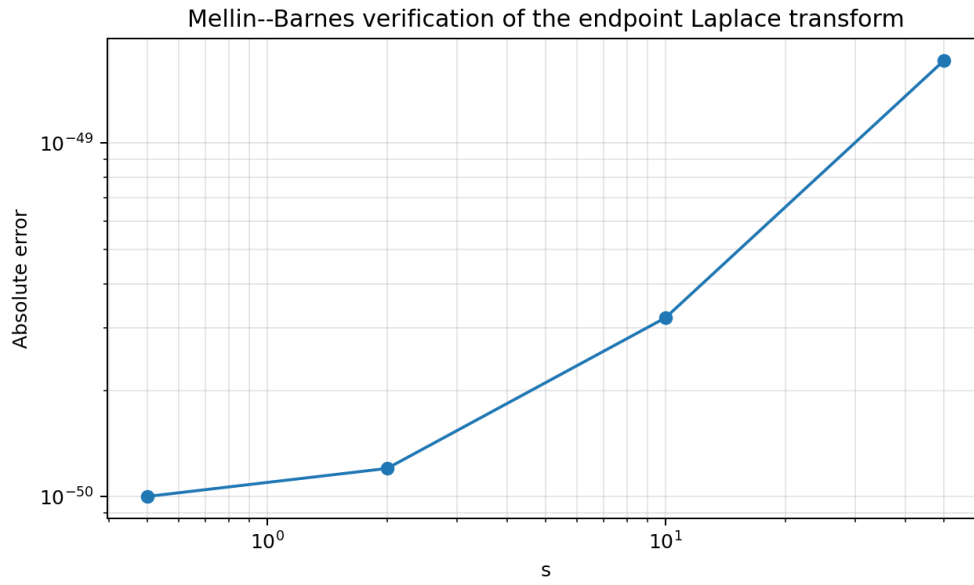


Figure 3: Absolute errors in the Mellin–Barnes check at 50-digit precision.

The near-machine-precision agreement is not evidence by itself, but it is a useful independent check of constants, scaling by  $2\pi$ , and the cancellation of the  $u = 1/2$  residue.

## 12 Further proved deductions and frontier conjectures

### 12.1 A unique critical point in every spectral gap

The logarithmic derivative can be written symmetrically as

$$\frac{\varphi'(t)}{\varphi(t)} = \sum_{m \geq 1} (1 + \nu_2(m)) \left( \frac{1}{t - 2\pi m} + \frac{1}{t + 2\pi m} \right). \quad (137)$$

On each interval  $(2\pi m, 2\pi(m + 1))$ , its derivative is strictly negative and its limits are  $+\infty$  and  $-\infty$ . Hence there is exactly one new zero of  $\varphi'$  in every positive spectral gap, in addition to the inherited



zeros at  $2\pi m$  of multiplicity  $\nu_2(m)$ . Denote this gap zero by  $c_m$ ; it is the unique solution of

$$\sum_{j \geq 1} \frac{1 + \nu_2(j)}{c_m^2 - (2\pi j)^2} = 0, \quad 2\pi m < c_m < 2\pi(m+1). \quad (138)$$

This resolvent characterization appears to be a useful starting point for the finer question below.

**Conjecture 12.1** (2-adic modulation of derivative-zero locations). *After subtracting a smooth expansion in descending powers of  $m$ , the normalized gap position*

$$\theta_m = \frac{c_m - 2\pi m}{2\pi} \in (0, 1) \quad (139)$$

*possesses a bounded nonconstant correction determined by the local 2-adic valuation pattern near  $m$ . More precisely, finite windows of  $(\nu_2(m-r), \dots, \nu_2(m+r))$  should determine successively accurate approximations to  $\theta_m$ .*

## 12.2 Fredholm minors for pointwise values

The determinant inversion formulas represent  $\text{up}(x)$  and  $F(x)$  by integrals of a scalar determinant. A more local operator formula would be stronger.

**Conjecture 12.2** (A pointwise Fredholm-minor model). *There exists a natural trace-class integral operator  $K_x$ , obtained from the self-similar Dirichlet tower by a boundary evaluation or rank-one coupling, such that  $\text{up}(x)$  on  $(-1, 1)$  is a Fredholm minor or a ratio of Fredholm determinants of  $K_x$ . The family can be chosen so that its Schur complement encodes the refinement equation (11).*

A promising construction is to replace the disconnected direct sum by a metric tree whose edge lengths decrease geometrically and to impose Kirchhoff conditions at the scale vertices. The direct-sum spectrum should then appear as the decoupled limit, while suitable boundary Green functions may recover pointwise values.

## 12.3 Operator saddle theory for the inverse Fabius function

The coefficient saddle (110) and the Bromwich phase derivative both involve

$$R(q) = \text{Tr}[B(I + qB)^{-1}] = \sum_{m \geq 1} \frac{1 + \nu_2(m)}{(2\pi m)^2 + q}. \quad (140)$$

Its complete monotonicity gives robust uniqueness and error bounds for positive saddles. This suggests an operator derivation of the lower-Lambert coordinate without first expanding the scalar product.

**Conjecture 12.3** (Resolvent-controlled inverse expansion). *The all-order endpoint expansion of  $F^{-1}(y)$  can be organized entirely in terms of  $R(q)$  and its derivatives at the unique positive saddle. After the leading Lambert inversion, every coefficient is a universal Bell polynomial in the spectral traces  $\text{Tr}[B^j(I + qB)^{-j}]$ , with the logarithmic-periodic component inherited from the poles of  $\zeta_{\mathcal{L}}$ .*

## 12.4 Beyond the classical dyadic tower

The family (62) turns exponent-sequence products into operators, but several arithmetic questions remain.

**Conjecture 12.4** (Schur positivity and arithmetic denominators). *For the classical spectrum, suitably scaled Schur values  $s_{\lambda}(\lambda_1, \lambda_2, \dots)$  have denominator bounds controlled by the hook lengths of  $\lambda$  and the factors  $2^{2j} - 1$  occurring in (98). These bounds should refine the known denominator theorems for  $F(2^{-n})$  when  $\lambda = (1^k)$ , and should extend to Pascal-weighted exponent sequences.*

For a geometric ratio  $q \neq 1/2$ , the spectral zeta denominator  $1 - q^{2s}$  moves the complex dimensions to an arithmetic progression with period  $|\log q|$ . This makes the operator family a controlled laboratory for separating features caused by compact convolution from those caused specifically by base two.

## 12.5 Exponentially small endpoint structure

**Conjecture 12.5** (Resurgent endpoint remainder). *After subtracting the quadratic logarithmic polynomial and the complete periodic Fourier series in (133), the remainder admits an exponentially small transseries whose actions are generated by the dyadic semigroup  $\{j2^r : j, r \in \mathbb{N}\}$ . Its Borel singularities are determined by the factors  $1 - e^{-s/2^n}$  in (32), and optimal truncation of the shifted Mellin–Barnes contour recovers that transseries.*

## 12.6 Formalization targets

The most accessible Lean targets are algebraic/operator statements that do not require the full endpoint saddle machinery:

1. construction of finite block operators  $B_N = \bigoplus_{n=1}^N 4^{-n}G$  and their determinant products;
2. passage to the trace-class limit and proof of (48);
3. Newton identities and the finite exact formula (105);
4. finite total positivity of the coefficient sequences, followed by stabilization of each fixed minor;
5. the resolvent identity (69) on compact subsets away from poles.

These steps would turn a substantial part of the new frontier package into a formal extension of the living repository.

## 13 Conclusion

The Fabius–Rvachev system has no shortage of formulas; the challenge is to organize them without confusing normalization, proof status, or historical priority. The corpus-known representation web begins with a random uniform series and reaches Thue–Morse splines, sinc and cosine products, 2-adic canonical products, Fourier and sampling integrals, exact dyadic  $q$ -arithmetic, Mellin harmonic sums, and Lambert-periodic endpoint asymptotics.

The specified operator

$$B = \bigoplus_{n \geq 1} 4^{-n} (-d^2/dx^2)_D^{-1}$$

adds a unifying layer. Its Fredholm determinant is the Fourier image; its inverse has 2-adic spectral degeneracies; its heat trace gives theta integrals; its resolvent gives partial fractions and saddle equations; its exterior powers are normalized even moments; its symmetric powers are moments of an exponential random series; and its gamma/Brownian laws explain complete monotonicity. The exact formula (105) connects reciprocal-dyadic Fabius values directly to this operator, while Pólya-frequency theory yields a new hierarchy of total-positivity and Turán inequalities. Finally, the Mellin–Barnes formula (127) identifies the endpoint logarithmic period with the complex dimensions of the same Dirichlet tower.

These links do not replace the existing  $q$ -binomial, Thue–Morse, Bell, Bernoulli, or Lambert descriptions. They explain why those descriptions coexist and supply new routes between them. The most promising next step is to develop Fredholm minors and operator saddles far enough to produce pointwise and inverse-function formulas, then to formalize the finite-block approximations and their trace-class limit.

## A Master table of representations

Table 4: A compact map of the principal representations.

| Object               | type of representation                                        | status             | reference    |
|----------------------|---------------------------------------------------------------|--------------------|--------------|
| $F$                  | CDF of $X = \sum 2^{-n} U_n$                                  | corpus-known       | (7)          |
| up                   | density of $Z = \sum 2^{-n} V_n$                              | corpus-known       | (7)          |
| up                   | infinite convolution of centered boxes                        | corpus-known       | (13)         |
| $u_N$                | finite Thue–Morse truncated-power spline                      | corpus-known       | (14)         |
| $\mathcal{F}$        | signed translate series and global self-differential equation | corpus-known       | (4), (5)     |
| $\varphi$            | dyadic sinc product                                           | corpus-known       | (15)         |
| $\varphi$            | multiplicity-weighted cosine product                          | corpus-known       | (16)         |
| $\Phi$               | 2-adic canonical product                                      | corpus-known       | (18)         |
| $\varphi$            | Fredholm determinant of a specified trace-class operator      | proved here        | (48)         |
| $\varphi$            | trace exponential involving $\zeta(2j)$                       | operator form      | (50)         |
| $\varphi'/\varphi$   | resolvent, valuation partial fractions, and cotangent scales  | new                |              |
| $\varphi(i\sqrt{q})$ | heat-kernel exponential and theta integral                    | operator bridge    | (69)         |
| $D_B^{-1/2}$         | Brownian-bridge quadratic-functional Laplace transform        | proved here        | (79)         |
| $D_B^{-1}$           | exponential/gamma random-series Laplace transform             | proved here        | (86)         |
| $e_n(B)$             | exterior-power trace and normalized even moment               | proved here        | (89), (90)   |
| $F(2^{-n})$          | positive finite exterior-power sum                            | proved here        | (97)         |
| $F(2^{-n})$          | Cauchy integral of a Fredholm determinant                     | proved here        | (105)        |
| $\{e_n\}$            | $\text{PF}_\infty$ , Toeplitz minors, Schur functions         | proved here        | (109)        |
| up                   | cosine Fourier determinant integral                           | proved here        | (113), (114) |
| $F$                  | sine Fourier determinant integral                             | operator form      | (119)        |
| $F$                  | Bromwich–Fredholm contour integral                            | new                |              |
| $L$                  | one-sided dyadic product                                      | operator form      | (120)        |
| $L$                  | heat/theta integral                                           | new                |              |
| $L$                  | Mellin–Barnes spectral-zeta integral                          | operator form      | (123)        |
| $\log L$             | quadratic logarithm plus periodic residue series              | new                |              |
| $F^{-1}$             | lower-Lambert and periodic saddle expansion                   | corpus-known       | (32)         |
|                      |                                                               | proved here        | (134), (135) |
|                      |                                                               | proved here        | (127)        |
|                      |                                                               | new derivation     | (133)        |
|                      |                                                               | of known structure |              |
|                      |                                                               | corpus-known       | (39), (40)   |

## B Proof details for the exterior-power dyadic formula

For completeness, the coefficient calculation behind [theorem 7.2](#) can be written in one line. From  $X = (1 + Z)/2$  and symmetry,

$$\begin{aligned}
 [z^n]A(z) &= \frac{\mathbb{E}X^n}{n!} = \frac{1}{2^n n!} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} \mathbb{E}Z^{2k} \\
 &= \frac{1}{2^n} \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{e_k(B)}{(n-2k)!}.
 \end{aligned}$$

Multiplication by  $2^{-\binom{n}{2}}$  proves (105). Equivalently,

$$A(2z) = e^z \varphi(iz) = e^z D_B(z^2), \quad (141)$$

so coefficient extraction immediately gives (106).

The first three nontrivial cases illustrate the mechanism:

$$F(1/4) = 2^{-3} \left( \frac{1}{2!} + e_1 \right) = \frac{1}{8} \left( \frac{1}{2} + \frac{1}{18} \right) = \frac{5}{72}, \quad (142)$$

$$F(1/8) = 2^{-6} \left( \frac{1}{3!} + e_1 \right) = \frac{1}{64} \left( \frac{1}{6} + \frac{1}{18} \right) = \frac{1}{288}, \quad (143)$$

$$F(1/16) = 2^{-10} \left( \frac{1}{4!} + \frac{e_1}{2!} + e_2 \right) = \frac{143}{2073600}. \quad (144)$$

## C Recursive corpus map and nonduplication checklist

The repository tree is unusually rich in revisions and parallel frontier reports. The audit was therefore conducted by lineage and theorem content rather than by comparing filenames alone. The following map records the groups that informed this report.

| Corpus group                           | material checked and its role                                                                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Living exposition                      | Definitions aligned with formal declarations; probability law; Fourier product and inversion; moments; exact dyadic arithmetic; source-of-truth normalizations.                               |
| Mathematical glossary                  | Notation reconciliation among $F$ , up, signed extensions, Thue–Morse data, $q$ -products, Bell and Bernoulli polynomials, and Lambert coordinates.                                           |
| Non-elementarity and Lean walkthroughs | Logical status, proof dependencies, and the boundary between formalized and research-only statements.                                                                                         |
| Consolidated frontier volume           | Endpoint Mellin analysis; logarithmic periodicity; inverse Fabius asymptotics; Fourier decay; local zero jets; arithmetic rays; half-integer sampling; integral and interpolation directions. |
| Exponent-sequence/Newton hierarchy     | Collapsed multiplicities, zero zeta, scalar spectral determinant, heat trace, arbitrary exponent sequences, Pascal weighting, and explicit warning that no operator spectrum was specified.   |
| Fourier-decay dossier family           | Sharp real-axis bounds, dyadic scale selection, stationary-scale envelopes, numerical validation, and nonduplication constraints for new Fourier formulas.                                    |
| Archived revisions                     | Historical development of exact dyadic, $q$ -binomial, Richardson, endpoint, and Thue–Morse results; used to prevent renaming an older result as new.                                         |
| Transcribed papers                     | Classical definitions, probability constructions, arithmetic of dyadic values, and comparison with the independent literature.                                                                |

The specific nonduplication checks most relevant to the new results were searches for operator-defining terms (Fredholm, trace class, Dirichlet Laplacian, Green operator), probability terms (Brownian bridge, gamma convolution), symmetric-function terms (exterior power, Schur, Pólya frequency, total positivity), and the exact spectral formula (105). The scalar zero-spectrum material was treated as prior art throughout.

## D Numerical archive contents

The ZIP archive contains the following reproducibility files.

### `numerical_experiments.py`

High-precision product, trace, exact rational, Turán, and Mellin–Barnes checks. The script is self-contained apart from the standard Python packages listed in its header.

### `fourier_product_trace_checks.csv`

Numerical comparison of (15) and (50).

### `exterior_coefficients.csv`

Exact  $e_n(B)$  and  $\mathbb{E}Z^{2n}$  through  $n = 18$ .

### `dyadic_exterior_values.csv`

Exact values from (105) through  $n = 15$ .

### `turan_inequalities.csv`

Ratios in (115) through  $n = 14$ .

### `mellin_barnes_checks.csv`

Direct-product and vertical-contour values for (127).

### `*.png`

The three figures embedded in section 11.

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